

Qns	Answer	Marks
6(a)(i)	Progressive wave is one where <u>energy is transferred in the direction of propagation of the wave.</u>	1
	Transverse wave is one where the <u>direction of oscillations is normal to the direction of energy propagation.</u>	1
6(a)(i)	Polarisation is where oscillations of a wave to one direction only, in the plane normal to the direction of energy propagation.	} 1
	Since the oscillations in a longitudinal wave is parallel to the direction of energy propagation, longitudinal wave cannot be polarised.	
6(b)(i)	Based on the amplitudes of the electricity field strength graphs, $A = A_o \cos \theta$ $16 = 9 \cos \theta$ $\theta = 56^\circ$	1 1
6(b)(ii)	$I \propto A^2$ $I = \cos^2(\theta)$ $\frac{I}{I_o} = \cos^2(\theta) = \left(\frac{9}{16}\right)^2 = 0.316$	1 1

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7(a)(i)	Loss in electric potential energy = gain in kinetic energy	